Dynamic Type Theory: Unifying Mathematics through the Principle of Efficient Novelty

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Abstract

The foundational tension in mathematics—whether truths are discovered in a Platonic realm or constructed through mental processes—has long defied resolution. Platonism explains objectivity but struggles with epistemic access, while Constructivism accounts for accessibility but falters on universality. We resolve this schism through Dynamic Type Theory (DTT), which augments Homotopy Type Theory (HoTT)—a paradox-free foundation unifying logic, computation, and geometry—with the Principle of Efficient Novelty (PEN). HoTT provides the static space of all potential mathematical structures as consistent types in a universe hierarchy. PEN supplies the dynamics: a constructive principle that generates, or 'actualizes,' a hierarchy of mathematical structures from first principles.

Central technical contribution. We show that DTT admits a *canonical, prefix-free enumeration of all logically consistent mathematical structures* (those definable in HoTT): fixing a single codebook yields a strict listing with no duplicates and no gaps; PEN then selects a deterministic subsequence of actualizations.

We axiomatize PEN via five principles: discrete logical substrate, effort as Kolmogorov complexity, novelty as total bit savings, deterministic maximization of the novelty-to-effort ratio for selection, and coherence/commitment for quantum-like superpositions and irreversibility. From these axioms, we rigorously derive key theorems: the golden ratio emerges as the unique thermodynamic fixed point for stable growth; effort and novelty values are uniquely determined without free parameters; and actualizations occur exclusively at Fibonacci-numbered steps in a deterministic genesis sequence. This sequence predicts the hierarchical emergence of mathematical structures—from universes and unit types to circles, manifolds, and the Einstein-Hilbert action—culminating in a 'Big Bang' of continuous fields, explaining the historical and logical order of mathematical development.

DTT transcends the Platonist–Constructivist divide: potential structures exist eternally in HoTT's logical space (objectivity), but actualize via constructive, efficiency-driven processes (accessibility). It resolves debates on excluded middle, choice, and infinity by contextualizing them within efficiency thresholds. By deriving its dynamics from principles of *algorithmic efficiency rather than new mathematical postulates*, DTT offers a new direction for the foundational program, portraying mathematics as the efficient actualization of logical possibilities.

1 Introduction: The Foundational Question

What does it mean for a mathematical object to exist? This deceptively simple question has profound implications for the nature of mathematical truth and knowledge. When we assert the existence of $\sqrt{2}$, or a group of order 60, or a continuous nowhere-differentiable function, what ontological claim are we making? The question sharpens when considering pathological objects: Does the set of all sets not containing themselves exist? Do inaccessible cardinals exist? Does a well-ordering of the reals exist? Mathematical history is marked by paradoxes arising from naive existence assumptions, from Russell's paradox to the independence of the continuum hypothesis.

Two philosophical traditions have dominated attempts to address these issues, each capturing partial truths while remaining incomplete.

Mathematical Platonism posits that mathematical objects exist eternally in an abstract realm, independent of human cognition or physical reality. Mathematical truths are discovered, not invented: the infinitude of primes held before humanity and persists regardless. This explains mathematics' objectivity, universality, and necessity—why theorems converge across cultures and why mathematical facts feel inevitable. Modern Platonists like Gödel invoke mathematical intuition as a perceptual faculty analogous to sensory perception, directed at abstract objects.

Yet Platonism faces a critical epistemic challenge, as articulated by Benacerraf: If mathematical objects reside in a causally isolated realm, how do physical beings access them? The success of mathematics seems to contradict its premises, requiring mysterious intellectual faculties.

Mathematical Constructivism, exemplified by Brouwer's intuitionism, counters that mathematical objects exist only through explicit mental construction. Existence proofs must furnish construction methods; the natural numbers arise from temporal intuition of succession. This resolves epistemic access—we know what we construct—and aligns with proof-as-construction practices, as in computer-verified theorems.

However, Constructivism struggles with uniformity: If mathematics is free creation, why the remarkable agreement among mathematicians? Why can't one construct contradictory objects? Objective constraints appear smuggled in without explanation, and rejecting principles like excluded middle seems to impoverish mathematics unnecessarily.

Formalism attempted a pragmatic escape: Mathematics as symbol manipulation, existence as consistency. Yet Gödel's incompleteness theorems revealed this inadequacy—systems cannot prove their consistency, and mathematics exceeds formal capture. Other syntheses—Kant's synthetic a priori, Frege-Russell logicism—faltered against non-Euclidean geometries and paradoxes.

Homotopy Type Theory (HoTT) marks a breakthrough, unifying logic (propositions as types), computation (proofs as programs), and geometry (types as spaces) in a paradox-free framework. HoTT's univalence axiom equates equivalent types, capturing mathematical practice while enabling computer verification. Yet HoTT remains static: It provides the space of all consistent types but no dynamics for selection. Why study groups over arbitrary operations? Why does mathematics develop hierarchically?

This paper introduces the Principle of Efficient Novelty (PEN) as the missing dynamics, yielding Dynamic Type Theory (DTT): HoTT's potential space governed by PEN's actualization process. PEN axiomatizes selection via algorithmic free energy minimization, where structures transition from potential to actual by balancing descriptive effort (Kolmogorov complexity) against generative novelty (algorithmic entropy).

DTT resolves the foundational tension: Potential structures exist eternally as HoTT types (Platonist objectivity), but actualize via constructive efficiency optimization (Constructivist accessibility). We formalize PEN through five axioms, deriving theorems on the golden ratio's emergence, metric uniqueness, and Fibonacci-paced genesis. This deterministic sequence predicts mathematics' emergence—from universes to manifolds—explaining order, fruitfulness, and physical applicability.

The paper proceeds as follows: Section 2 traces the historical dialectic. Section 3 details HoTT's structural foundation and limitations. Section 4 axiomatizes PEN. Section 5 proves core theorems. Section 6 derives the genesis sequence. Section 7 resolves debates with examples. Section 8 explores implications and open questions. DTT completes foundations, revealing mathematics as efficient actualization of logical possibilities.

2 The Historical Dialectic

The tension between discovered and constructed mathematics defines the central fault line in mathematical philosophy. To appreciate how Dynamic Type Theory resolves this, we must examine the evolution of this dialectic from ancient origins through modern crises to contemporary synthesis attempts. Each position and attempted resolution highlights the need for a framework

that distinguishes potential from actual existence while incorporating efficiency dynamics.

2.1 The Platonic Thesis

Mathematical Platonism originates in Pythagorean insights that numerical ratios underlie reality, such as musical harmonies (octave: 2:1, fifth: 3:2). Plato formalized this in the *Republic*, positing mathematical objects in an eternal, ideal realm more real than the physical. Knowledge involves recollection of these forms; a drawn triangle participates imperfectly in the ideal triangle.

This view accounts for key features:

1. **Objectivity**: Truths independent of opinion, e.g., no rational π.
2. **Universality**: Independent discoveries (Pythagorean theorem in multiple ancient cultures).
3. **Necessity**: Impossibility of alternative truths (e.g., $2 + 2 = 5$).
4. **Discovery Experience**: Intuitions like Poincaré's sudden solutions.

Gödel refined this, arguing mathematical intuition perceives abstract objects, fallible yet reliable like sensory perception.

Formally, Platonism implies a realm where propositions have determinate truth values independent of proof. Yet Benacerraf's dilemma persists: Causal isolation precludes knowledge. In HoTT terms, if types exist platonically, how do computational proofs (terms inhabiting types) bridge to physical minds?

2.2 The Constructivist Antithesis

Constructivism arose amid early 20th-century paradoxes, with Brouwer arguing crises stem from treating mathematics as pre-existing rather than constructed. Mathematics begins with temporal intuition: succession yields naturals via finite operations.

Consequences include:

1. **Rejected Excluded Middle**: Asserting $P \vee \neg P$ requires constructing P or its refutation.
2. **Potential Infinity**: Naturals as process, rejecting completed $\mathbb{N}$.
3. **Proof as Meaning**: "$\exists x\, P(x)$" demands a constructive x and proof $P(x)$.
4. **Revised Theorems**: No constructive intermediate value theorem without additional assumptions.

This solves epistemology: Knowledge via construction. It fits practices like computer proofs (e.g., four-color theorem).

Yet puzzles remain: Uniformity across minds? Objective constraints (no square circle)? In HoTT, constructivism aligns with proofs-as-terms, but lacks explanation for why certain constructions (e.g., reals via cuts) prevail universally.

2.3 Failed Syntheses

Attempts to bridge the divide incorporate insights but fall short.

Kant's Critical Philosophy: Mathematics as synthetic a priori, rooted in cognitive structures (space/time as intuitions). Explains applicability and a priori knowledge, but non-Euclidean geometries refute rigidity—alternative geometries are conceivable and physically applicable.

Formally, Kant's framework predicts Euclidean necessity, contradicted by hyperbolic geometry's consistency.

Formalism (Hilbert): Mathematics as formal games; existence as consistency. Program: Prove consistency finitarily. Gödel's theorems shatter this: Arithmetic systems incomplete, unprovable consistency. Formalism ignores content and fruitfulness distinctions among consistent systems.

In DTT terms, formalism lacks dynamics to select fruitful axioms.

Logicism (Frege-Russell): Reduce mathematics to logic. Numbers as logical classes; truths as analytic. Russell's paradox necessitates types, but restrictions seem arbitrary. Gödel shows arithmetic requires non-logical axioms.

These teach: Cognitive structure needed but flexible (contra Kant); consistency insufficient without dynamics (contra formalism); logic-mathematics links deep but incomplete (contra logicism).

A successful synthesis requires:

- Objectivity without inaccessible realms.
- Construction without subjectivism.
- Formality with meaningful content.
- Explanation of physical effectiveness.
- Paradox resolution without arbitrariness.

HoTT provides structure: Paradox-free via types, unifying logic/computation/geometry. PEN adds dynamics: Efficiency selects among possibilities. DTT synthesizes: Eternal potentials (HoTT types) actualized efficiently (PEN), transcending the dialectic.

3 HoTT: The Structural Foundation

Homotopy Type Theory (HoTT) represents a paradigm shift in mathematical foundations, synthesizing insights from logic, computer science, and topology into a unified framework. By interpreting types as spaces, terms as points, and proofs as paths, HoTT provides a paradox-free arena for mathematics that is both constructive and geometrically intuitive. This section elucidates HoTT's core elements, achievements, and limitations, establishing it as the static substrate upon which PEN's dynamics operate. All definitions and theorems herein draw from the standard HoTT literature [**?**].

3.1 The Unification in HoTT

HoTT builds upon Martin-Löf's intensional type theory, extended with homotopy-theoretic interpretations. The foundational correspondence is the Curry-Howard isomorphism: - Propositions are types. - Proofs are terms inhabiting those types.

This evolves into a richer structure where: - Logical implication $A \to B$ is the function type. - Conjunction $A \wedge B$ is the product type $A \times B$. - Disjunction $A \vee B$ is the sum type $A + B$. - Existence $\exists x : A \,.\, P(x)$ is the dependent sum $\sum_{x:A} P(x)$. - Universality $\forall x : A \,.\, P(x)$ is the dependent product $\prod_{x:A} P(x)$.

The homotopy layer introduces identity types $\mathrm{Id}_A(x, y)$ (or $x =_A y$), representing proofs of equality between terms $x, y : A$. Traditionally propositional, these gain geometric structure: - Reflexivity: $\mathrm{refl}_x : x = x$ (constant path). - Symmetry: If $p : x = y$, then $p^{-1} : y = x$ (path reversal). - Transitivity: If $p : x = y$ and $q : y = z$, then $p \cdot q : x = z$ (path concatenation).

Higher identities (paths between paths) yield ∞-groupoids, with types as spaces.

The univalence axiom elevates this:

$$\mathrm{UA} : (A \simeq B) \simeq (A =_{\mathcal{U}} B), \tag{1}$$

where $\simeq$ denotes equivalence (bijection with coherent inverses), and $\mathcal{U}$ is a universe type. Equivalent types are identical, formalizing mathematical practice of treating isomorphic structures interchangeably.

3.2 Key Constructs in HoTT

HoTT's power stems from its universe hierarchy and inductive definitions.

Definition (Universes)**.** HoTT features a cumulative hierarchy of universes $\mathcal{U}_0 : \mathcal{U}_1 : \mathcal{U}_2 : \cdots$, where each $\mathcal{U}_i$ is a type whose terms are types of level $< i$. This avoids paradoxes like Russell's by stratification.

Definition (Inductive Types)**.** Types are defined inductively, e.g., natural numbers:

$$\begin{gathered} 0 : \mathbb{N}, \\ \succ : \mathbb{N} \to \mathbb{N}, \\ \mathrm{ind}_{\mathbb{N}} : \prod_{P:\mathbb{N}\to\mathcal{U}} P(0) \to \left(\prod_{n:\mathbb{N}} P(n) \to P(\succ n) \right) \to \prod_{n:\mathbb{N}} P(n). \end{gathered}$$

Definition (Circle S^1 as HIT)**.**

$$\begin{gathered} \mathrm{base} : S^1, \\ \mathrm{loop} : \mathrm{base} = \mathrm{base}, \\ \mathrm{ind}_{S^1} : \cdots \quad \text{(eliminator with computation rules).} \end{gathered}$$

This yields $\pi_1(S^1) \simeq \mathbb{Z}$.

3.3 Achievements of HoTT

HoTT advances foundations in several dimensions:

1. **Paradox-Free**: Hierarchical universes prevent self-referential paradoxes; no type of all types.
2. **Constructive**: Proofs yield explicit constructions; $\exists$ requires a witness term.
3. **Computational**: Terms are executable programs; normalization ensures decidable equality for finite types.
4. **Higher-Dimensional**: Captures ∞-categories natively, e.g., homotopies as 2-paths.
5. **Internal Logic**: Metatheory internalized; theorems about types are types.
6. **Unification**: Handles discrete (e.g., $\mathbb{N}$), continuous (e.g., $\mathbb{R}$ via HITs), and higher structures uniformly.

For instance, S^1 generates rich homotopy: winding numbers correspond to integers, enabling topology within type theory.

3.4 Limitations of HoTT

Despite these strengths, HoTT lacks mechanisms to explain mathematical practice:

1. **Selection Problem**: All consistent types exist equally; no principle favors groups over arbitrary magmas.
2. **Actualization Gap**: No distinction between potential (consistent) and actual (fundamental) structures.
3. **Development Mystery**: Cannot predict hierarchical emergence (e.g., groups before categories).
4. **Construction Order**: Infinite possibilities, but no guidance on pursuit order.

To formalize:

Lemma 1 (Potential Space in HoTT)**.** *For any consistent type specification T (inductive schema without contradictions), there exists a universe level i such that $T : \mathcal{U}_i$. Thus, HoTT provides the* ***formal space*** *for all potential mathematical structures.*

Proof. By universe cumulativity and inductive closure, consistent definitions inhabit some $\mathcal{U}_i$. Contradictions (e.g., Russell-type) fail formation rules. □

This lemma positions HoTT as the "potential space" for PEN: all possibilities exist logically, but actualization requires construction.

HoTT thus provides the canvas—vast, paradox-free, unified—but lacks the brush. PEN supplies the constructive evolution, yielding DTT: proven mathematics as actualized potentials.

4 Abstract Syntax Tree for Mathematics: A Canonical Indexation of Mathematical Structures

To apply the dynamics of PEN, we must index the potential structures within HoTT. We achieve this by defining a canonical codebook that allows any mathematical structure to be uniquely represented as a binary string. We fix a concrete, prefix-free binary codebook L_{HoTT} for well-formed HoTT terms (terms modulo definitional equality), together with a length-only parser $\mathsf{parseLen}$ that recognizes a single top-level codeword. An *abstract syntax tree* (AST) is the standard tree-structured representation of a formal language expression, where internal nodes denote operations and leaves denote atomic symbols. The encoder $\mathsf{encodeAST} : \mathrm{AST} \to \{0,1\}^*$ is compositional (tags + length-prefixed payloads) and the parser only needs sizes of subblocks; both are verified in Lean (Appendix A).

Theorem A — Abstract Syntax Trees for Mathematics. *There exists a prefix-free codebook L_{HoTT} and a length-only parser* $\mathsf{parseLen}$ *such that:*

(a) ***Correctness:*** *For every well-formed HoTT term p,*

$$\mathsf{parseLen}(\mathsf{encodeAST}(p)) = \mathrm{some}(|\mathsf{encodeAST}(p)|).$$

(b) ***Prefix-freedom:*** *If* $\mathsf{encodeAST}(p)$ *is a prefix of a byte string b and b is also a valid codeword, then $b = \mathsf{encodeAST}(p)$.*

(c) ***Completeness:*** *Every finite, well-typed HoTT term has a unique codeword in L_{HoTT} (modulo definitional equality).*

Consequently, L_{HoTT} induces a canonical, effective indexation $\mathrm{Index} : \mathbb{N} \to \mathsf{AST}/\equiv$ *of all potential mathematical structures, compatible with Kraft's inequality and unique up to computable recoding.*

Proof sketch. (a) and (b) follow from a length-only parser spec with two local properties: (A) $\mathsf{parseLen}$ returns the exact size on encodings, and (B) prefix-stability for valid codewords. These are discharged by byte-level lemmas (size additivity of concatenation, 1-byte tags, and length-prefixed blocks). (c) follows because the encoder covers all constructors of the HoTT syntax and is injective modulo definitional equality. Full details are mechanized in Lean; see Appendix A for the statements `parseLen_encode` and `encode_prefix_free`. □

Remark 1. This theorem turns the "space of potentials" into a concrete, prefix-free tree with exact bit-lengths.

4.1 Codebook and Canonical Indexing

We fix a canonical prefix-free language L_{HoTT} for well-formed HoTT objects (types, terms, and contexts). Let

$$\mathsf{CodeRange} \;=\; \{\, \mathsf{encode}_{\mathrm{AST}}(p) \mid p \text{ a well-formed HoTT object} \,\} \subseteq \{0,1\}^*.$$

By prefix-freeness there is a length-lexicographic bijection

$$\mathrm{Index} : \mathbb{N} \xrightarrow{\simeq} \mathsf{CodeRange}, \qquad n \mapsto \text{the } n\text{th codeword by increasing length, then lex order.}$$

Remark 2 (Static indexing vs. sequential ordering). The map Index is a *static* naming of logically possible structures. It does not imply any sequential ordering of these structures.

5 PEN: Actualization Dynamics

6 The Principle of Efficient Novelty

Intuition. Mathematics grows by converting promising *potentials* into *actual* structures. At each tick of time, candidate structures compete on a single score: *efficiency* $\rho = \nu/\kappa$—future savings (ν) per unit of construction effort (κ). Two simple ideas drive the dynamics: (i) only look one small ring out when estimating future savings; (ii) remember recent performance and demand that new winners clear an adaptive bar. This creates a self-organizing trajectory: small tools enable medium ones; mediums reveal higher-level patterns; the process repeats.

Definition 1 (Computer-Aided Declaration (CAD) System). We model Mathematical Reality $\mathcal{R}$ within a formal system called CAD. $\mathcal{R}$ is represented by a formal Context Γ, a finite collection of realized Atomic Declarations (‘AtomicDecl‘). The system evolves via ‘Derivations‘ (sequences of declarations) drawn from a finite action alphabet $\mathcal{A}$. A validity predicate ‘isValidInContext(decl,)‘ enforces structural dependencies (e.g., a term requires its type former; a specific affordance may require a prerequisite organizing term).

Axiom 1 (Discrete Evolution and Hierarchical Integration)**.** Mathematical Reality $\mathcal{R}$ is a cumulative sequence of formal CTT contexts. $\mathcal{R}$ evolves through discrete logical time $\tau \in \mathbb{N}$, beginning from the empty set $\mathcal{R}(0) = \emptyset$. The evolution is strictly cumulative:

$$\mathcal{R}(\tau - 1) \subseteq \mathcal{R}(\tau)$$

When a new structure X_n is actualized at time τ_n, the system enters an integration period $\Delta_n = \tau_{n+1} - \tau_n$. The duration of this period is determined by the following principles:

1. **Integration Cost:** The duration equals the organizational complexity of the domain that X_n organizes:

$$\Delta_n = \kappa(\text{domain organized by } X_n)$$

2. **Hierarchical Synthesis:** For $n \geq 2$, the domain organized by a new structure X_n is the conceptual synthesis of the domains organized by the two preceding structures, X_{n-1} and X_{n-2}. Its complexity is therefore the sum of their complexities:

$$\kappa(\text{domain of } X_n) = \kappa(\text{domain of } X_{n-1}) + \kappa(\text{domain of } X_{n-2})$$

The cumulative nature ensures that mathematical knowledge is never lost. The principle of Hierarchical Synthesis formalizes the observation that each major mathematical insight serves to unify and build upon the two most recent conceptual foundations. This creates an

intrinsic growth pattern that optimally balances memory (the structures of τ_{n-1} and τ_{n-2}) with momentum (the new possibilities opened by X_n), forcing the system to evolve at the golden ratio. Time, in this framework, is not merely a counter but the computational cost of this hierarchical synthesis.

Definition 2 (Effort κ as Minimal Derivation Length). Given a base context B and a target structure (package) X, the raw effort $\kappa_{raw}(X|B)$ is the length of the shortest valid derivation $D \subseteq \mathcal{A}$ that installs X on B. This minimum is found via Iterative Deepening Depth-First Search (IDDFS).

The final Effort $\kappa(X|B)$ incorporates *Coherence Discounts* $\Delta_{coh}(X)$ for recognized self-organizing bundles (e.g., Pi/Sigma duality, sealed HITs), reflecting reduced essential complexity.

$$\kappa(X|B) = \kappa_{raw}(X|B) - \Delta_{coh}(X)$$

Definition 3 (Novelty Gauge H_ν and Truncated Effort). The Novelty Gauge is a fixed, small horizon used to measure the immediate impact of a structure, set to $H_\nu = 2$. The truncated effort $\kappa_H(Y|B)$ is the effort $\kappa(Y|B)$ if it is $\leq H$, and $H + 1$ otherwise.

Definition 4 (Immediate Frontier $\mathcal{F}$). The Immediate Frontier $\mathcal{F}(X|B)$ is the set of concepts $Y \in \mathcal{A}$ that become accessible in a single step after realizing X.

$$\mathcal{F}(X|B) = \{Y \in \mathcal{A} \setminus X \mid \kappa(Y|B \cup X) = 1\}$$

Definition 5 (Schema Keying and Capacity). A Schema Keying function $K(Y)$ maps declarations to 'FrontierKey' classes. 1. **Coarse Graining vs. Exactness:** It balances grouping similar concepts (e.g., general terms of the same host) with maintaining distinctness for unique organizational concepts (e.g., path structure terms). 2. **Endogeneity:** We define Endogenous Keys $\mathcal{K}_{endo}(X)$ as the keys associated with the internal machinery of X. 3. **Capacity:** Each key k has a Capacity $C(k)$ (typically 2 or 3), limiting the maximum novelty contribution from that schema class, reflecting its intrinsic complexity.

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Axiom 3 Novelty as Algorithmic Compression

The Novelty $\nu(X|B)$ of a structure X relative to basis B measures the total reduction in effort required to access the Immediate Frontier $\mathcal{F}(X|B)$. It is calculated using the fixed Novelty Gauge $H_\nu = 2$.

Axiom 2 (Novelty Calculation (AIT Formulation))**.** The calculation of $\nu(X|B)$ proceeds as follows:

1. Effort Reduction Calculation: For each concept Y on the Immediate Frontier $\mathcal{F}(X|B)$, we calculate the raw reduction in effort achieved by realizing X.

$$\Delta_{raw}(Y) = \kappa_{H_\nu}(Y|B) - \kappa(Y|B \cup X)$$

2. Apply Capacity Constraints: The effective contribution $\Delta(Y)$ is the raw reduction capped by the capacity of its schema key $C(K(Y))$.

$$\Delta(Y) = \min(\Delta_{raw}(Y), C(K(Y)))$$

3. Schema Keying and Exclusion: The frontier $\mathcal{F}$ is partitioned into equivalence classes based on the Schema Keying function K.

* **Exclusion:** We filter the frontier to exclude concepts whose keys are endogenous to X.

$$\mathcal{F}_{ext} = \{Y \in \mathcal{F}(X|B) \mid K(Y) \notin \mathcal{K}_{endo}(X)\}$$

* **Maximal Gain per Key:** For each remaining (external) Schema Key K_i, we calculate the maximal effort reduction achieved within that class:

$$\Delta_{max}(K_i) = \max_{Y \in \mathcal{F}_{ext}: K(Y) = K_i} \Delta(Y)$$

4. Novelty Summation: The core novelty is the sum of the maximal effort reductions over all external schema classes.

$$\nu_{core}(X|B) = \sum_{K_i} \Delta_{max}(K_i)$$

The total novelty includes a bonus for packages that introduce new abstraction layers:

$$\nu(X|B) = \nu_{core}(X|B) + \text{Bonus}(X)$$

where $\text{Bonus}(X) = 1$ if X consists entirely of Type Formers, and 0 otherwise.

6.1 Axiom 4: Selection by Minimal Overshoot

The selection mechanism determines which structures get realized based on their efficiency relative to a dynamic threshold that rises with accumulated mathematical sophistication.

Axiom 3 (Competition and Selection)**.** At each tick τ, the algorithm selects at most one structure for realization through the following process:

Define the Global Resonance Bar:

$$\text{Bar}(\tau) = \Phi(\tau) \cdot \Omega(\tau - 1)$$

where:

- $\Omega(\tau - 1) = \frac{\sum_{t<\tau} \nu(t)}{\sum_{t<\tau} \kappa(t)}$ is the historical average efficiency
- $\Phi(\tau) = \frac{\tau}{\tau_{-1}}$ is the momentum factor (ratio of current time to time of last realization)

Admissibility Criteria: A structure X is admissible at time τ if:

1. $\kappa(X \mid \mathcal{R}(\tau - 1)) \leq H$ (within current effort horizon)
2. $\text{Level}(X) \leq \max_{Y \in \mathcal{R}(\tau-1)} \text{Level}(Y) + 1$ (respects universe hierarchy)
3. X can be constructed from structures in $\mathcal{R}(\tau - 1)$

Selection Rule:

1. Calculate efficiency $\rho(X) = \nu(X)/\kappa(X)$ for all admissible candidates
2. If $\max_X \rho(X) < \text{Bar}(\tau)$: idle (no realization), increment horizon $H \to H + 1$
3. Otherwise: realize the candidate with minimal positive overshoot:

$$\text{Winner} = \arg \min_{X:\rho(X) \geq \text{Bar}(\tau)} (\rho(X) - \text{Bar}(\tau))$$

4. After realization: reset horizon $H \to 2$

Tie-Breaking: If multiple candidates achieve identical minimal overshoot, select those with minimal κ. If ties persist, realize all tied candidates in superposition, weighted by their number of distinct minimal derivations.

The Global Resonance Bar serves as a quality control mechanism that prevents premature or incremental realizations. The historical average Ω ensures that new structures must be increasingly efficient as mathematics matures, while the momentum factor Φ prevents indefinite stagnation by gradually lowering the threshold during idle periods.

The minimal overshoot principle is crucial: it ensures that mathematics develops through conceptually coherent steps rather than jumping to highly efficient but premature abstractions. This creates a "conceptual ratchet" where each realization is the natural next step given current mathematical reality.

The horizon mechanism (H) controls the search space for candidates. Starting at $H = 2$ after each realization forces the next innovation to be lean and directly build on recent progress. When no candidates clear the bar, the horizon expands, allowing more complex structures to be considered.

Together, these four axioms create a self-organizing system where mathematical structures emerge not randomly but in order of optimal conceptual compression, producing a growth pattern that balances exploration of new structures with integration of existing ones.

6.2 The Genesis Sequence

Theorem G — Genesis Sequence. (Genesis Sequence) *Under Axioms 1–5, with the fixed canonical codebook L_{HoTT}, the PEN dynamics determine a* unique *actualization history $\{\mathcal{R}(\tau)\}_{\tau\in\mathbb{N}}$ such that:*

(i) Actualization occurs only at Fibonacci ticks

$$\tau \in \{1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597, ..\}.$$

(ii) At each such τ, exactly one structure X_τ is realized, with integer effort $\kappa(\tau)$ and novelty $\nu(\tau)$ as in Table **??***; in particular*

$$\frac{\nu(\tau)}{\kappa(\tau)} \;>\; \phi \cdot \Omega(\tau - 1),$$

where $\phi = (1 + \sqrt{5})/2$ and $\Omega(\tau - 1) = \big(\sum_{t<\tau} \nu(t)\big)/\big(\sum_{t<\tau} \kappa(t)\big)$.

(iii) No other candidate definable from $\mathcal{R}(\tau-1)$ both (a) exceeds the effective bar $\phi\cdot\Omega(\tau-1)$ and (b) strictly surpasses the realized ratio at that tick.

Table 1: The first 16 Fibonacci actualizations in the Genesis Sequence.

τ	*Structure*	ν	κ	ν/κ	*Bar* $= \phi \cdot \Omega(\tau - 1)$
1	*Universe* $\mathcal{U}_0 : \mathcal{U}_1$	1	2	0.500	—
2	*Unit type* $\mathbf{1}$	1	1	1.000	0.809
3	*Witness* $\star : \mathbf{1}$	2	1	2.000	1.079
5	Π/Σ *types*	5	3	1.667	1.618
8	*Circle* S^1	7	3	2.333	2.080
13	*Smooth manifolds*	8	3	2.667	2.589
21	*Higher sphere* S^2	10	3	3.333	2.987
34	3-*Sphere* $S^3 \cong \mathrm{SU}(2)$	18	5	3.600	3.438
55	*Hopf fibration* $S^3 \to S^2$	17	4	4.250	4.007
89	*Generic Lie groups*	9	2	4.500	4.468
144	*Differentiable atlases*	19	4	4.750	4.674
233	*Principal bundle connections*	26	5	5.200	5.068
377	*Curvature tensors*	34	6	5.667	5.528
610	*Metric + orthonormal frame bundle*	43	7	6.143	6.048
987	*Hilbert functional on Riemannian metrics*	60	9	6.667	6.604
1597	*First smooth function space;* $\mathbb{R}$	150	8	18.750	7.253

Implication: Generative Mathematics. Because DTT specifies a deterministic actualization dynamics over HoTT's space of possibilities, it functions as an algorithm for the automatic construction of mathematics. Unlike brute-force enumeration, PEN filters candidates by efficiency, producing an ever-expanding body of mathematics that is not only internally consistent but also hierarchically coherent and fruitful. This suggests that the mathematics generated by DTT is likely to be "meaningful and effective" in the sense of yielding structures that simplify, unify, and ultimately prove useful in scientific theories. In this light, DTT can be interpreted as a constructive foundation not just for existing mathematics, but for mathematics yet to be discovered.